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United States Patent	12406125
Kind Code	B2
Date of Patent	September 02, 2025
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Integrated circuit layout including standard cells and method to form the same

Abstract

A method for forming an integrated circuit layout including at least two standard cells having different cell heights is disclosed. The standard cells respectively have a well boundary to divide a PMOS region and an NMOS region. The standard cells are abutted side by side along their side edges in a way that the well boundaries of the cells are aligned along the row direction. The power rail and the ground rail of one of the standard cells are shifted to align and connect to the power rail and the ground rail of the other one of the standard cells.

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Appl. No.: 17/868770

Filed: July 19, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230097189 A1	Mar. 30, 2023

Foreign Application Priority Data

TW	110135987	Sep. 28, 2021
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Related U.S. Application Data

Publication Classification

Int. Cl.: G06F30/392 (20200101); G06F30/30 (20200101)

U.S. Cl.:

CPC G06F30/392 (20200101); G06F30/30 (20200101);

Field of Classification Search

CPC: G06F (30/392); G06F (30/30)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. application Ser. No. 17/517,642, filed on Nov. 2, 2021. The content of the application

is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to the field of integrated circuits. More particularly, the present invention relates to a method for forming an integrated circuit layout using standard cells and an integrated circuit layout obtained therefrom.

2. Description of the Prior Art

(2) As the needs for better performance and various applications are increased, the integrated circuits (ICs) has become more and more complicated and usually includes hundreds of thousands or millions of logic gates. In the industry, to facilitate circuit design process, circuit units of specific logic functions commonly used in an integrated circuit are usually designed into standard cells with logic gates, such as NAND cells, NOR cells, d-flip-flop cells, latch cells, I/O cells, OP amplifier cells, ADC cells, DAC cells. After verifying the manufacturability for mass production of the standard cells, standard cell libraries including the verified standard cells may be authorized to the chip designers to construct functional circuitries by electronic design automation (EDA) tools such as logic simulators, logic synthesizers, and automatic placement/routing tools. A typical circuit design process usually involves specifying the functionality of the circuit using a hardware programming language, synthesizing/mapping the resulting circuit description into basic logic gates of standard cell libraries, placing and routing physical layouts based on the gate netlist, and finally verifying proper connectivity and functionality of the layout. In this way, a complex and large integrated circuit layout may be correctly constructed automatically within a short period of time.

(3) A standard cell library associated with a specific logic function may include hundreds of standard cells that can be selectively combined to design a larger circuit. A standard cell is usually laid out relative to a grid defined by horizontal and vertical tracks. The number of horizontal tracks defines the height of the cell (also referred to as cell height or track height). The number of vertical tracks defines the width of the cell (also referred to as cell width or track width). Conventional standard cell libraries are comprised of cells having the same height to enable cells of the cell library to be readily combined to create larger circuits. The widths of standard cells in the library may vary.

(4) In advanced technology, in order to optimize area efficiency, speed and power consumption of the synthesized integrated circuit, a cell library may be provided with standard cells having devices laid out in different dimensions and therefore having different cell heights. However, intermixing of the standard cells having different cell heights in a same routing block are constrained by reduced efficiency of the synthesis tool due to pattern irregularity. Besides, manufacturing process window may also be impacted. To resolve the above problems, a common approach taken currently is to put the standard cells with different cell heights into different monolithic routing blocks and electrically connect the standard cells by metal interconnections. However, this has adversely limited the design flexibility. The extended length of the metal interconnections may also increase the power consumption during operation of the integrated circuit.

(5) What is therefore needed in the field is a method able to effectively produce an integrated circuit layout including intermixing of standard cells having different cell heights, and an integrated circuit layout obtained therefrom.

SUMMARY OF THE INVENTION

(6) The present invention is directed to provide a method for forming an integrated circuit layout including mixed-height standard cells retrieved from a same standard cell library or from different cell libraries, and an integrated circuit layout obtained therefrom. The present invention is able improve design flexibility, synthesis efficiency, and inline process manufacturability.

(7) One embodiment of the present invention provides a method for forming an integrated circuit

layout including the following steps. First, a first standard cell and a second standard cell of a same function and having different cell heights are selected. The first standard cell and the second standard cell respectively include a power rail, a ground rail, and a well boundary between the power rail and the ground rail extending in parallel along a first direction. Two active regions of opposite conductivity types are arranged between the power rail and the ground rail and at two sides of the well boundary. A gate line extends along a second direction and intersects the two active regions, wherein the first direction and the second direction are perpendicular. Following, the first standard cell and the second standard cell are abutted side by side to form a temporary placement in a way that the well boundaries of the first standard cell and the second standard cell are aligned along the first direction. After that, based on the temporary placement, a layout is generated by shifting the power rail and the ground rail of the second standard cell along the second direction to align and connect to the power rail and the ground rail of the first standard cell.

(8) Another embodiment of the present invention provides an integrated circuit layout, which includes a power rail and a ground rail extending in parallel along a first direction. A first standard cell and a second standard cell of a same function are placed between the power rail and the ground rail and abutted to each other side by side. The first standard cell and the second standard cell respectively include an upper edge, a lower edge, a well boundary between the upper edge and the lower edge extending in parallel along the first direction, two active regions of opposite conductivity types at two sides of the well boundary, and a gate line extending between the upper edge and the lower edge along a second direction and intersecting the two active regions. The first direction and the second direction are perpendicular. A first cell height between the upper edge and lower edge of the first standard cell and a second cell height between the upper edge and lower edge of the second standard cell are different. The well boundaries of the first standard cell and the second standard cell are aligned along the first direction.

(9) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings are schematic drawings and included to provide a further understanding of the embodiments, and are incorporated in and constitute a part of this specification. The drawings illustrate some of the embodiments and, together with the description, serve to explain their principles. Relative dimensions and proportions of parts of the drawings have been shown exaggerated or reduced in size and are not necessarily drawn to scale, for the sake of clarity and convenience in the drawings. The same reference signs are generally used to refer to corresponding or similar features in modified and different embodiments.
- (2) FIG. 1 is a schematic plan view illustrating how standard cells are placed within rows of an integrated circuit layout according to an embodiment of the present invention.
- (3) FIG. 2 is a flow chart illustrating a method for forming an integrated circuit layout according to an embodiment of the present invention.
- (4) FIG. 3, FIG. 4 and FIG. 5 are plan views illustrating the steps of the flow chart shown in FIG. 2.
- (5) FIG. 6 is a schematic plan view of an integrated circuit layout according to an embodiment of the present invention.
- (6) FIG. 7 is a schematic plan view of an integrated circuit layout according to an embodiment of the present invention.
- (7) FIG. 8 is a schematic plan view of an integrated circuit layout according to an embodiment of the present invention.

(8) FIG. 9 is a circuit diagram of the integrated circuit shown in FIG. 5.

(9) FIG. 10 depicts an electronic design automation (EDA) environment used to perform the method shown in FIG. 2.

DETAILED DESCRIPTION

(10) To provide a better understanding of the present invention to those of ordinary skill in the art, several exemplary embodiments of the present invention will be detailed as follows, with reference to the accompanying drawings using numbered elements to elaborate the contents and effects to be achieved. The accompanying drawings are included to provide a further understanding of the embodiments, and are incorporated in and constitute a part of this specification. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments may be utilized and that structural, logical and electrical changes may be made without departing from the spirit and scope of the present invention.

(11) For the convenience of illustrating the spatial relationships of the features, the first direction X and the second direction Y which are perpendicular to each other are shown in the drawing. In addition to the orientation shown in the drawings, other orientations (for example, rotated by 90 degrees or other directions) of the present invention may also be explained by the spatially relative descriptions in the specification. The term “cell width” or “track width” refer to a width of a standard cell taken along the first direction X and between two side edges of an abutment box (or abutment area) of the cell. The term “cell height” refer to a height of a standard cell taken along the second direction Y and between an upper edge and a lower edge of the abutment box, and may be referred to as “track height”. The width of a power rail or a ground rail refers to a width taken along the second direction Y, and may be referred to as “track width”. The cell height and the cell width may be described as a predetermined number of tracks, such as 7T, 8T or 9T.

(12) It should be understood that the number of the gate line of the standard cells in the embodiments of the present invention are only examples, and may be modified in other embodiments without departing from the scope of the present invention. The number of the gate line of the present invention may be single or plural.

(13) FIG. 1 is a schematic plan view of a portion of an integrated circuit layout 10, illustrating how standard cells having different cell heights are placed within a row according to an embodiment of the present invention. Four rows are exemplarily shown in FIG. 1. The rows respectively run along the first direction X and are butted to each other along the row boundaries BN. The first direction X is also referred to as the row direction. The rows have a fixed row height RH, which means that the row boundaries BN of the integrated circuit layout 10 are equally spaced. Standard cells are placed within the rows and tied to voltage rails (such as power rails and ground rails) that run continuously through the integrated circuit layout 10 along the row boundaries BN. It is noteworthy that the constituent standard cells of the integrated circuit layout 10 may have different cell heights, and the row height RH is equal to the maximum cell height of the standard cells to be placed within the rows.

(14) For example, as shown in FIG. 1, the integrated circuit layout 10 may include standard cells 12 and 14 abutted side by side within a row. The standard cells 12 and 14 may be selected from different cell libraries or a mixed-height cell library associated with a specific logic function. The standard cell 12 includes a abutment box, which includes an upper edge 12a, a lower edge 12b, and the two side edges 12c, wherein a cell height H12 of the first standard cell 12 is defined by the distance between the upper edge 12a and the lower edge 12b, and a cell width W12 of the first standard cell 12 is defined by the distance between the two side edges 12c. The standard cell 14 includes a abutment box, which includes an upper edge 14a, a lower edge 14b, and the two side edges 14c, wherein a cell height H14 of the standard cell 14 is defined by the distance between the upper edge 14a and the lower edge 14b, and a cell width W14 of the standard cell 14 is defined by the distance between the two side edges 14c. The standard cells 12 and 14 are abutted by overlapping the side edges 12c and 14c to reduce useless areas between them. The standard cells 12

and **14** have a same logic function (such as AND gates, NAND gates, inverters, OR gates, NOR gates, or flip flops, but is not limited thereto) but different cell heights for the purpose of laying out functional components in different dimensions for different electrical performances. According to an embodiment of the present invention, the cell height **H12** of the standard cell **12** may be larger than the cell height **H14** of the standard cell **14**, and equal to the row height **RH**. Therefore, the upper edge **12a** and the lower edge **12b** of the standard cell **12** are overlapped with the row boundaries **BN**, while the upper edges **14a** and the lower edge **14b** of the standard cell **14** are spaced from the row boundaries **BN** by same or different distances. Functional components of the standard cells **12** and **14** laid within the abutment boxes are tied to a same power rail and/or a same ground rail running along the row boundaries **BN** at the top and bottom of the row.

(15) Another standard cell **16** may be placed in a row adjacent to the row of the standard cells **12** and **14**. The standard cell **16** may be selected from another cell library, the same cell library of the standard cells **12** or **14**, or the mixed-height cell library of the standard cells **12** and **14**. The standard cell **16** may have an abutment box and a cell height and a cell width. When the cell height of the standard cell **16** is equal to the row height **RH**, the upper edge and the lower edge of the abutment box of the standard cell **16** may overlap the row boundaries **BN**, and the abutment box of the standard cell **16** is abutted to the lower edge **12b** of the abutment box of the standard cell **12**. When the cell height of the standard cell **16** is smaller than the row height **RH** (similar to the standard cell **14**), the upper edge and the lower edge of the abutment box of the standard cell **16** would be spaced from the row boundaries **BN** by same or different distances. In this case, the abutment boxes of the standard cells **12** and **16** would be separated. According to an embodiment of the present invention, the functional components of the standard cells **12** and **16** are tied to a same power rail (or a same ground rail, depending the orientation of the cells) along the row boundary **BN** between them. In this way, space efficiency of the integrated circuit layout **10** may be improved.

(16) Please refer to FIG. 2 to FIG. 5 and FIG. 9. FIG. 2 is a flow chart illustrating a method for forming an integrated circuit layout using standard cells having different cell heights according to an embodiment of the present invention. FIG. 3 to FIG. 5 are plan views showing how to abut an exemplary first standard cell Cell-A and an exemplary second standard cell Cell-B having different cell heights to form an integrated circuit layout through the method shown in FIG. 2. FIG. 9 is a circuit diagram of the integrated circuit shown in FIG. 5. The method shown in FIG. 2 may be carried out in an electronic design automation (EDA) environment.

(17) At step **22**, a first standard cell and a second standard cell are selected based on a netlist of an integrated circuit. The first standard cell and second standard cell have the same logic function but have different cell heights and different electrical performances. For example, as shown in FIG. 3, the first standard cell Cell-A and the second standard cells Cell-B may be inverters. In detail, the first standard cell Cell-A includes an abutment box (the box delineated with bold dashed line) having an upper edge **A1** and a lower edge **A2** extending in parallel along the first direction **X**, and two side edges **A3** extending in parallel along the second direction **Y**. The first standard cell Cell-A has a cell height **H1** defined by the upper edge **A1** and the lower edge **A2**, and a cell width **W1** defined by the two side edges **A3**. A centerline **114** runs through the center of the abutment box along the first direction **X**, and divides the abutment box into two equal parts. In other words, the distances from the centerline **114** to the upper edge **A1** and the lower edge **A2** are the same. Two active regions **120p** and **120n** of opposite conductivity types are respectively arranged in the two parts of the abutment box. A gate line **130** extends along the second direction **Y** and intersects the two active regions **120p** and **120n**. Two dummy gate lines **132** parallel to the gate line **130** are arranged along the two side edges **A3** at two sides of the two active regions **120p** and **120n**. According to an embodiment of the present invention, the gate line **130** and the dummy gate lines **132** have a same length along the second direction **Y**, and ends of the gate line **130** and the dummy gate lines **132** are flush with each other along the first direction **X**. A well region **116** completely

overlaps the upper part of the abutment box and the active region **120p**. As shown in FIG. 3, the well boundary **116a** of the well region **116** may be overlapped with the centerline **114** of the abutment box. According to an embodiment of the present invention, the active region **120p** is p-type, and the active region **120n** is n-type. The intersecting area of the gate line **130** and the active region **120p** forms a p-type metal oxide semiconductor transistor (PMOS). The intersecting area of the gate line **130** and the active region **120n** forms an n-type metal oxide semiconductor transistor (NMOS). The portions of the active regions **120p** and **120n** at the left side (the side near the conductive connectors **142** and **152**) of the gate line **130** are source regions S of the PMOS and the NMOS. The portions of the active regions **120p** and **120n** at the right side of the gate line **130** are drain regions D of the PMOS and the NMOS. The well region **116** defines an n-well where the p-type active region **120p** is to be formed in a p-type substrate during fabrication of the integrated circuit.

(18) The first standard cell Cell-A further includes a power rail **140** and a ground rail **150** respectively arranged on the upper edge A1 and the lower edge A2 along the second direction Y. The widths of the power rail **140** and the ground rail **150** may be the same or different. According to an embodiment of the present invention, the centerline **140a** of the power rail **140** that divides the power rail **140** into two equal parts may be completely overlapped with the upper edge A1. The centerline **150a** of the ground rail **150** that divides the ground rail **150** into two equal parts may be completely overlapped with the lower edge A2.

(19) The first standard cell Cell-A further includes conductive connectors and contact plugs to interconnect the transistors and the power rail and the ground rail to enable the functionality of the first standard cell Cell-A. In detail, conductive connectors **142** and **152** are arranged at a same side of the gate line **130**, respectively connect to the power rail **140** and the ground rail **150** and partially overlap the source regions S of the active regions **120p** and **120n**. A conductive connector **162a** is arranged at the other side of the gate line **130** and partially overlaps the drain regions D of the active regions **120p** and **120n**. A conductive connector **162b** is arranged at the same side as the conductive connectors **142** and **152** and has a protrusion partially overlapping the middle portion of the gate line **130**. A plurality of contacts plugs **118** are provided to electrically connect the source region S of the active region **120p** to the conductive connector **142** and the power rail **140**, the source region S of the active region **120n** to the conductive connector **152** and the ground rail **150**, the gate line **130** to the conductive connector **162b**, and the drain regions D of the active regions **120p** and **120n** to the conductive connector **162a**. According to an embodiment of the present invention, the power rail **140**, the ground rail **150**, and the conductive connectors **142**, **152**, **162a**, **162b** are laid on a same layout layer, such as metal-1 layer. According to some embodiments of the present invention, the first standard cell Cell-A is a high performance cell having larger dynamic current and faster speed.

(20) Same as the first standard cell Cell-A, the second standard cell Cell-B includes an abutment box (the box delineated with bold dashed line) having an upper edge B1, a lower edge B2, two side edges A3, a cell height H2, and a cell width W2. A centerline **214** runs through the center of the abutment box of the second standard cell Cell-B. A p-type active region **220p** and an n-type active region **220n** are arranged at two sides of the centerline **214**. A gate line **230** intersects the active regions **220p** and **220n**, forming a PMOS and an NMOS, respectively. Two dummy gate lines **232** are at two sides of the two active regions **120p** and **120n**. The gate line **230** and the dummy gate lines **232** may have a same length along the second direction Y. Line ends of the gate line **230** and the dummy gate lines **232** may be flush with each other along the first direction. A well region **216** completely overlaps the upper part of the abutment box of the second standard cell Cell-B, and may have a well boundary **216a** overlapped with the centerline **214**. A power rail **240** and a ground rail **250** respectively arranged on the upper edge B1 and the lower edge B2. The centerline **240a** of the power rail **240** that divides the power rail **240** into two equal parts may be completely overlapped with the upper edge B1. The centerline **250a** of the ground rail **250** that divides the ground rail **250**

into two equal parts may be completely overlapped with the lower edge B2. The width of the power rail **240** of the second standard cell Cell-B may be the same as the width of the power rail **140** of the first standard cell Cell-A. The width of the ground rail **250** of the second standard cell Cell-B may be the same as the width of the ground rail **150** of the first standard cell Cell-A. Conductive connectors **242**, **252**, **262a**, **262b** and contact plugs **218** are provided to interconnect the transistors and the power rail **240** and the ground rail **250** to enable the functionality of the first standard cell Cell-B. Other detailed descriptions of the components of the second standard cell Cell-B may be referred to previous descriptions of the first standard cell Cell-A, and are not explained herein for the sake of brevity. According to some embodiments of the present invention, the second standard cell Cell-B is a low power cell having better area efficiency and lower power leakage. Along the second direction Y, a width of the active region **220p**, a width of the active region **220n**, and a length of the gate line **230** of the second standard cell Cell-B are respectively smaller than a width of the active region **220p**, a width of the active region **220n**, and a length of the gate line **130** of the first standard cell Cell-A. The cell height H2 of the second standard cell Cell-B is smaller than the cell height H1 of the first standard cell Cell-A.

(21) Subsequently, at step **24**, the first standard cell and the second standard cell are abutted side by side to form a temporary placement **10A**. As shown in FIG. **4**, the first standard cell Cell-A and the second standard cell Cell-B are abutted in a way that the side edges A3, B3 of the first standard cell Cell-A and the second standard cell Cell-B are overlapped, and the well boundary **116a** of the first standard cell Cell-A and the well boundary **216a** of the second standard cell Cell-B are aligned along the first direction X. According to an embodiment of the present invention, the dummy gate line **132** and the dummy gate line **232** on the overlapped side edges A3, B3 are overlapped and combined to form a shared dummy gate line **332**. According to an embodiment of the present invention, the dummy gate line **332** is spaced from the gate line **130** and the gate line **230** by a same pitch. According to an embodiment of the present invention the dummy gate line **332** may have a same length as the gate line **130** along the second direction Y. Line ends of the dummy gate line **332** and the gate line **130** are flush with each other along the first direction X.

(22) Subsequently, at step **26**, an integrated circuit layout is formed based on the temporary placement. As shown in FIG. **4** and FIG. **5**, after abutting the first standard cell Cell-A and the second standard cell Cell-B, the power rail **240** and the ground rail **250** of the second standard cell Cell-B are recognized and shifted along the second direction Y respectively by a distance P1 and a distance P2 (shown in FIG. **4**) until the centerline **240a'** of the shifted power rail **240'** is aligned with the centerline **140a** of the power rail **140**, and the centerline **250a'** of the shifted ground rail **250'** is aligned with the centerline **150a** of the ground rail **150**. Therefore, the power rail **240'** and ground rail **250'** of the second standard cell Cell-B are respectively connect to the power rail **140** and ground rail **150** of the first standard cell Cell-A smoothly, such that continuous power **340** and ground rails **350** are obtained. According to an embodiment of the present invention, the distance P1 and the distance P2 are the same. The conductive connectors **242** and **252** are elongated along the second direction Y to keep connecting to the power rail **240** and the ground rail **250** as the power rail **240** and the ground rail **250** are shifted, while the contact plugs **118** remain in their original positions. According to an embodiment of the present invention, the length L1 of the elongated conductive connector **242** and the length L2 of the elongated conductive connector **252** may be the same. The length L3 of the conductive connector **142** may be equal to or different from the length L1 and the length L2 of the conductive connectors **242** and **252**. After the step **26**, an integrated circuit layout **10B** provided by the present invention is obtained.

(23) Subsequently, at step **28**, the connectivity and functionality of the integrated circuit layout **10B** are verified. After that, the integrated circuit layout **10B** is output to a set of photomasks used in a manufacturing process to form an integrated circuit chip.

(24) Please refer to FIG. **9**, which is a circuit diagram corresponding to the integrated circuit layout **10B** as shown in FIG. **5**. Cell-A and Cell-B are electrically coupled in a cascade manner, and

respectively include a PMOS T1 and an NMOS T2. In each cell, the gates of the PMOS T1 and the NMOS T2 are connected to each other and form an input terminal of the cell which is electrically coupled to an input voltage V_{in} . The drain regions D of the PMOS T1 and the NMOS T2 are coupled with each other and form an output terminal of the cell which is coupled to an output voltage V_{out} . The source region S of the PMOS T1 is coupled to the power rail V_{dd} , and the source region S of the NMOS T2 is coupled to the ground rail V_{ss} . The output terminal of Cell-B is the input terminal of Cell-A. The power rail V_{dd} is coupled to a high voltage or an operation voltage. The ground rail V_{ss} is coupled to a low voltage, a reference voltage, or a ground voltage.

(25) It is noteworthy that in the embodiment shown in FIG. 3 to FIG. 5, the distances P1 and P2 are the same since the centerlines 114 and 214 of the first standard cell Cell-A and the second standard cell Cell-B are aligned. The distance D1 between the power rail 240' and the upper edge B1 of the second standard cell Cell-B may be the same as the distance D2 between the ground rail 250' and the lower edge B2 of the second standard cell Cell-B, and is larger than the distance D3 between the power rail 140 and the upper edge A1 of the first standard cell Cell-A. In some embodiments, edges a1 and b1 of the active regions 120p and 220p of the same conductivity type (such as p-type) adjacent and parallel to the well boundaries 116a and 216a may be aligned along the first direction X (for example, along the line I'). In some embodiments, edges a2 and b2 of the active regions 120n and 220n of the same conductivity type (such as n-type) adjacent and parallel to the well boundaries 116a and 216a may be aligned along the first direction X (for example, along the line II).

(26) The following description will detail the different embodiments of the present invention. To simplify the description, identical components in each of the following embodiments are marked with identical symbols. For making it easier to understand the differences between the embodiments, the following description will detail the dissimilarities among different embodiments and the identical features will not be redundantly described.

(27) In some embodiments, the n-type active region and the p-type active region of the standard cell may have different well enclosure/space rules. The well boundary may not overlap the centerline of the standard cell. For example, please refer to FIG. 6, which is a schematic plan view of an integrated circuit layout 10C according to an embodiment of the present invention. The integrated circuit layout 10C shown in FIG. 6 is substantially the same as the integrated circuit layout 10B shown in FIG. 5 except that the well boundaries 116a and 216a are not overlapped with the centerlines 114 and 214 of the first standard cell Cell-A and the second standard cell Cell-B. More specifically, the well boundary 116a is at a side of the centerline 114 closer to the active region 120p of the first standard cell Cell-A and extends along the first direction X. The well boundary 216a is at a side of the centerline 214 closer to the active region 220p of the second standard cell Cell-B and extends along the first direction X. In this embodiment, the distance between the boundary 116a and the centerline 114 and the distance between the boundary 216a and the centerline 214 may be the same. Therefore, after abutting the first standard cell Cell-A and the second standard cell Cell-B in the way that the boundaries 116a and 216a are aligned along the first direction X, the centerline 114 and the centerline 214 may also be aligned along the first direction X. Similar to the integrated circuit layout 10B shown in FIG. 5, the length L1 of the elongated conductive connector 242 and the length L2 of the elongated conductive connector 252 may be the same. The length L3 of the conductive connector 142 may be equal to or different from the length L1 and the length L2 of the conductive connectors 242 and 252.

(28) Please refer to FIG. 7, which is a schematic plan view of an integrated circuit layout 10D according to an embodiment of the present invention. The integrated circuit layout 10D shown in FIG. 7 is substantially the same as the integrated circuit layout 10B shown in FIG. 5 except that the well boundary 216a is not overlapped with the centerline 214 of the second standard cell Cell-B. More specifically, the well boundary 216a is at a side of the centerline 214 closer to the active region 220p of the second standard cell Cell-B and extends along the first direction X. The well

boundary **116a** is overlapped with the centerline **114** of the first standard cell Cell-A. After abutting the first standard cell Cell-A and the second standard cell Cell-B in the way that the boundaries **116a** and **216a** are aligned along the first direction X, the centerline **114** and the centerline **214** are offset along the first direction X. In this embodiment, the shifted distance P1 (shown in FIG. 4) of the power rail **240** is larger than the shifted distance P2 (shown in FIG. 4) of the ground rail **250**, and the length L1 of the elongated conductive connector **242** is larger than the length L2 of the elongated conductive connector **252**. The length L1 may also be larger than the length L3 of the conductive connector **142**.

(29) Please refer to FIG. 8, which is a schematic plan view of an integrated circuit layout **10E** according to an embodiment of the present invention. The integrated circuit layout **10E** shown in FIG. 8 is substantially the same as the integrated circuit layout **10B** shown in FIG. 5 except that the well boundary **116a** is not overlapped with the centerline **114** of the first standard cell Cell-A. More specifically, the well boundary **116a** is at a side of the centerline **114** closer to the active region **120p** of the first standard cell Cell-B and extends along the first direction X. The well boundary **216a** is overlapped with the centerline **214** of the second standard cell Cell-B. After abutting the first standard cell Cell-A and the second standard cell Cell-B in the way that the boundaries **116a** and **216a** are aligned along the first direction X, the centerline **114** and the centerline **214** are offset along the first direction X. In this embodiment, the shifted distance P1 (shown in FIG. 4) of the power rail **240** is smaller than the shifted distance P2 (shown in FIG. 4) of the ground rail **250**, and the length L1 of the elongated conductive connector **242** is smaller than the length L2 of the elongated conductive connector **252**. The length L1 may also be smaller than the length L3 of the conductive connector **142**.

(30) Please refer to FIG. 10, which depicts an electronic design automation (EDA) environment used to perform the method shown in FIG. 2 according to an embodiment of the present invention. The environment **100** includes specification tools **102**, synthesis tools **104**, placement/routing tools **106**, verification tools **108**, and a standard cell library portfolio **110**. During the design process, the functionality of the chip is specified in a specification tool **102** using a standard hardware programming language such as verilog. The resulting circuit description is synthesized/mapped into the basic gates of standard cells retrieved from at least a cell library from the standard cell library portfolio **110** by using one or more synthesis tools **104** such as DesignCompiler (produced by Synopsys, Inc.). The resulting gate netlist is then placed and routed using placement/routing tools **106** such as BlastFusion (produced by Magma, Inc.). Finally, the connectivity and functionality of the integrated circuit are verified using a verification tool **108**.

(31) In conclusion, the present invention provides a method to generate an integrated circuit layout including mixed-height standard cells. The cells are abutted on their side edges and, critically, have their well boundaries aligned along the row direction. Following, the voltage rails of the cell having a smaller cell height are shifted to align and connect to the voltage rails of the cell having a larger cell height, and thereby continuous voltage rails running along the row boundaries to tie the standard cells are obtained. The synthesis methodology of the present invention effectively facilitates the synthesis process, and the integrated circuit layout obtained therefrom may avoid rule violations such as violations of the well enclosure/space rules. The use of these mixed-height standard cells may produce circuits having optimized area, speed and power efficiency.

(32) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. An integrated circuit layout, comprising: a power rail and a ground rail extending in parallel along a first direction; and a first standard cell and a second standard cell abutted to each other side by side between the power rail and the ground rail, which respectively comprising: an upper edge, a lower edge, and a well boundary between the upper edge and the lower edge extending in parallel along the first direction; two active regions of opposite conductivity types at two sides of the well boundary; and a gate line extending between the upper edge and the lower edge along a second direction and intersecting the two active regions, wherein the first direction and the second direction are perpendicular; wherein a first cell height between the upper edge and lower edge of the first cell and a second cell height between the upper edge and lower edge of the second standard cell are different, the well boundaries of the first standard cell and the second standard cell are aligned along the first direction.
 2. The integrated circuit layout according to claim 1, wherein a centerline between the upper edge and the lower edge of the first standard cell and a centerline between the upper edge and the lower edge of the second standard cell are aligned along the first direction.
 3. The integrated circuit layout according to claim 2, wherein the centerline and the well boundary of each of the first standard cell and the second standard cell are overlapped.
 4. The integrated circuit layout according to claim 2, wherein the centerline and the well boundary of each of the first standard cell and the second standard cell are not overlapped.
 5. The integrated circuit layout according to claim 1, wherein a distance between the power rail and the upper edge of the second standard cell is larger than a distance between the power rail and the upper edge of the first standard cell.
 6. The integrated circuit layout according to claim 1, wherein a centerline between the upper edge and the lower edge of the first standard cell and a centerline between the upper edge and the lower edge of the second standard cell are offset along the first direction, and a distance between the power rail and the upper edge of the second standard cell is different from a distance between the ground rail and the lower edge of the second standard cell.
 7. The integrated circuit layout according to claim 1, wherein edges of the active regions of a same conductivity type of the first standard cell and the second standard cell adjacent and parallel to the well boundary are aligned along the first direction.
 8. The integrated circuit layout according to claim 1, wherein edges of the active regions of a same conductivity type of the first standard cell and the second standard cell adjacent to the well boundary are offset along the first direction.
 9. The integrated circuit layout according to claim 8, further comprising: a first conductive connector extending from an edge of the power rail along the second direction a first length to partially overlap one of the active regions of the first standard cell; and a second conductive connector extending from the edge of the power rail a second length to partially overlap one of the active regions of the second standard cell, wherein the first length and the second length are different.
 10. The integrated circuit layout according to claim 1, further comprising: a first dummy gate line at a side of the first standard cell opposite to the second standard cell; a second dummy gate line at a side of the second standard cell opposite to the first standard cell; and a third dummy gate line between the first standard cell and the second standard cell, wherein the first dummy gate line and the third dummy gate line have line ends flush with line ends of the gate line of the first standard cell, the second dummy gate line has line ends flush with line ends of the gate line of the second standard cell.
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